
To Decide or not to Decide: Decision Making and Decision Avoidance in Critical Incidents

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The authors' research area concerns the specialist domain of police decision making and leadership. It is firmly rooted within a criminal justice context; in particular, it focuses on police responses to and investigation of critical incidents. Despite the specialist focus, generic literature on decision making has much to offer and some of the findings we discuss in this chapter are as relevant to lawyers as any other professional or lay person. On the other hand, a criminal justice context has distinct features and we argue that context has an overwhelming impact on decision making so factors that are specific to a criminal justice context must be considered. Both police officers and barristers may claim that they prepare a case for prosecution. Even though each is ostensibly carrying out the same task, their different environments, aims, procedures and constraints will have different influences on their decision making.

In this chapter, we shall first review different decision-making theories and their relative merits and weaknesses; next we shall outline some of the factors that can have an impact on decision making¹ from those factors that can influence an individual (e.g. how emotion can affect decision making), to a consideration of the organizational and political culture in which professionals work and how this broader context can also affect decision making in critical incidents. Examples throughout this chapter will be criminal justice related and they mostly refer to decision making in critical incidents. There are some examples indicating the application and relevance of a particular point to the field of law but, limited by space we invite, indeed encourage, the reader to consider throughout how the issues we discuss in this chapter might apply to their own particular field of expertise.

¹ To avoid unwieldy repetition, the term 'decision making' is sometimes used interchangeably with 'decision avoidance'.

INTRODUCTION – THE ALL-IMPORTANT CONTEXT

To be or not to be. That is the question: Whether 'tis nobler in the mind to suffer The slings and arrows of outrageous fortune, Or to take arms against a sea of troubles, And by opposing end them? (*Hamlet*)

'To be or not to be. That is the question' – perhaps the most famous quotation of all and for anyone who has read or seen Shakespeare's play, Hamlet takes a pretty long time to make his mind up. The consequences of his decision or nondecision will have an impact on Hamlet and other sundry characters, but all the actors can wash off the greasepaint and go home after he has finally made a decision.

By contrast, professionals involved in managing critical incidents are often under intense time pressure to make decisions and the consequences of getting it wrong can mean life or death for others, sometimes great numbers of ordinary members of the public. Critical incidents constitute a very particular context where immense pressure is brought to bear upon the professionals involved, both pressure of time and the magnitude of the potential consequences if they make the wrong decision.

As we have moved into the twenty-first century, cultural change has altered the broader context in which critical incident managers work. For example in a policing context, the government white paper, *Policing a New Century: A Blueprint for Reform* (2001) states that relations with communities are an integral part of the current government's agenda for change. The police as a whole will need to adapt their policies and procedures in response to political demands; in turn, there are implications for decision making in critical incidents.

Relations with communities are increasingly filtered through the lens of the media. The new century has seen the advent of 24-hour news media with a voracious appetite for immediate stories. Managers of critical incidents have to respond to the media and be aware of the very real fact that the common thrust of such news gathering is 'Who's to Blame?' Equally real is the prospect of a public inquiry that may well have the same 'Who's to Blame?' agenda, should things go wrong. Beyond that, an increasingly litigious culture awaits with at least a widespread perception that more claims for negligence are being made.

How, then, are decisions made in such daunting circumstances? Decision making has been studied by psychologists for decades. However, the influence of organizations and the broader cultural context surrounding critical incidents is under-researched. Equally important (but much neglected in the literature) is how and when people fail to make decisions at all. Now they are the interesting questions.

TRADITIONAL, NATURALISTIC AND PRAGMATIC APPROACHES TO DECISION MAKING

Traditional Decision-Making Theory

Traditional Decision-Making Theory (TDM) cast humans as principally rational beings. Faced with several alternative choices, decision makers conduct a thorough search for information and calculate the subjective expected utility (SEU) of each option. The option with the highest SEU is then chosen—decision made. Based largely on traditional

experimental studies in the laboratory, TDM has contributed a good deal to the understanding of the cognitive processes involved when decision makers are faced with choices. Careful consideration of options may obtain in some situations but decision making is a much more complex process than the early models suggested.

Heuristics and Biases

Seminal work by Kahneman and Tversky (1973) indicated that humans are not always so rational and an exhaustive search for information is not always conducted. Rather, individuals rely on heuristics as cognitive short cuts to reduce the time taken to explore endless alternatives. These two authors also identified several biases that influence decision making, particularly in conditions of uncertainty. For example, decision makers may make judgements on the basis of how easily they can call to mind other examples they deem to be relevant to the decision in question. Termed the 'Availability Bias', decision makers will incorporate the most salient, 'available' information into the decision-making process and neglect other information that may be relevant or even crucial. (See Kahneman and Tversky (1973) for a fuller discussion of biases and heuristics.) In decision making, heuristics are important in reducing cognitive load and processing time, but there is potential for decision errors because individuals' judgements are susceptible to well-documented biases and an individual's heuristics may be idiosyncratic. A legal issue, arising from this, has not received the attention it deserves. To avoid being negligent, a professional must, amongst other requirements, make decisions consistent with those that a responsible body of coprofessionals would make (*Bolam v. Friern HMC* ([1957] 4 All ER 118)). Is the test normative, how those professionals ought to have decided (i.e. unaffected by availability and other biases), or descriptive, how those professionals, being human and fallible, would have decided? Fortunately, or otherwise, lawyers have not paid sufficient attention to decision research, for the manner in which decisions are taken, not just their content, might provide fruitful pickings for compensation claims.

Researchers learned, then, that decision makers are not entirely rational and are affected by a lifetime's baggage of experiences and biases. These biases and heuristics were understood to exist but the focus was still very much on the individual decision maker in the here and now, rather than an exploration of the social context in which all humans exist and the ways in which it comes to influence decision makers over the long term.

Thus, one of the limitations of TDM is that it did not consider context. Important though it is to understand cognitive processes, humans do not exist in a social vacuum and context necessarily influences the way that decisions are made, from different situations, through the context, that is, the group or team of people who make decisions together to the organizational context that imposes constraints on decision makers.

A second, related limitation of TDM was methodological. The methodological strength of laboratory studies is the control the researcher has over the experiment. The weakness is that laboratory studies can have limited ecological validity; that is, their applicability beyond the lab to the 'real-world' that is far more complex and far less controllable. TDM offers us only a limited and imperfect idea about how decisions are made. For example, university students in a laboratory study take their time to decide which hypothetical car dealer has the best offer. A critical incident manager must decide in moments whether or which workers to send to the scene of a shooting. The consequences of that decision will be

the lead news story before they arrive home from work. The two situations are simply not comparable. TDM was not invalid, just incomplete and decision-making theory required further development.

Naturalistic Decision Making Theory

Naturalistic decision-making theory (NDM) addressed some of the limitations of TDM with a greater focus on what decision makers actually do in 'natural' conditions in the real-world. It advocates the study of practitioners making decisions in real-life, familiar contexts. In essence, NDM researchers argue that the starting point for research is to investigate what *is* done rather than what *ought* to be done. There are disagreements between different researchers with claims that TDM research actually does investigate 'what is' done by experts and there is also debate about whether 'what is' can be divorced from 'what ought'. Similarly, other researchers argue that such distinctions are somewhat contrived. In a bid to draw distinctions between different schools of thought in decision-making theory, it is possible that a false dichotomy has been created.

No matter what the starting point for decision-making research (traditional laboratory or naturalistic experts in the field), ultimately all research can uncover systematic weaknesses. Therefore, there must be a normative framework within which to operate, one that includes optimal solutions or at the very least, the framework should include satisfactory solutions. Prescription, thereafter, is unavoidable if the research is to have any practical value; in other words, what *ought* to be done necessarily follows from what *is* done. NDM researchers may criticize TDM researchers because they prefer the creation of 'ideal world' conditions in a laboratory instead of looking at 'what *is* done' by expert decision makers, but NDM researchers must at some point embrace the goal of an 'ideal world' if they are to improve the decision making they find practitioners using in their day-to-day professional lives. In this respect, to avoid prescription is to relegate one's research to the merely descriptive; description can be an extremely useful starting point, but it not a useful end point for research. (See LeBoeuf and Shafir, 2001; Lipshitz *et al.*, 2001.)

Detailed distinctions and different emphases are beyond the scope of this chapter, but some substantive points hold. NDM research gives a greater emphasis to expertise and how decisions are made in familiar, real-world contexts. These contexts are dynamic and multi-faceted and typically occur under great time constraints; indeed, it can be difficult even to define with any great precision what the decision problem is. TDM, on the other hand, deals with well-defined decision-making problems, focuses on the individual decision maker in a fairly stable unchanging context and is not usually subject to time constraints. Although the decision making under investigation may be mundane by comparison, researchers can be sure which variables they are investigating and controlling. The issues are also reflected in the key legal distinction between subjective and objective *mens rea*. The defendant's behaviour may be adjudged as, objectively, very dangerous. If the legal test is objective then they will be guilty of a crime that can be committed recklessly, for example manslaughter. However, if the law requires the defendant to have been subjectively aware of wrong-doing, their misperceptions, misunderstandings and general confusion at the time of making the decision leading to harm, could allow them to escape liability (see Chapter 9, this volume.). Such issues are at the heart of the review of the law of homicide, in England and Wales, being undertaken, at the time of writing, by the Law Commission (2005).

An NDM perspective, then, recognized the context to a greater degree in that it did not view decision making as a discrete event uninfluenced by other factors. It took greater account of the situation in which decision makers made their choices and what it lacked in controllability found in lab-based studies, it compensated for in the greater ecological validity of field based work.

Recognition Primed Decision Making

A prototypical model within NDM is Recognition Primed Decision Making (RPDM) (Klein, 1993). Expertise in the field was used as a starting point for research and the subsequent development of Klein's model where he analysed the decision-making processes employed by experienced fire-fighting commanders. The model was devised from self-reports of the strategies used by commanders.

It transpired that expert decision makers can assess the situation and, far from considering and weighing alternative options (as predicted by SEU theory), they can simply decide to take action based on the first option they generate. Based on experience, they can assess the typicality of a situation and act appropriately.

Where a situation is not typical and there is some uncertainty, decision makers often rely on a story building strategy and generate a narrative about the events that might have led up to the situation. They can also use forward based reasoning and construct mental simulations to consider hypothetically how a specific action might unfold.

RPDM sheds light on 'what is done' by expert commanders dealing with fire-fighting incidents and indicated how decision makers can cope with extreme time pressure in emergency situations. The model has since had empirical support in other arenas. For example, Randal and Pugh (1996) compared expert and novice electronic warfare technicians from US navy ships, whose tasks include monitoring the number of hostile ships that appear on radar screens. They found that experts placed greater emphasis on assessing the situation whereas novices spent more time considering different courses of action. In accordance with the RPDM model, having assessed the situation thoroughly, the experts generated one option only; that option usually became their course of action. RPDM can be thought of as pattern matching using 'If > then' rules (sometimes referred to as 'productions')—*If (situation) A, then (action) X*. Unsurprisingly, the experts also made more correct decisions than novices. Kobus, Proctor and Holste (2001) also found that experts take longer to assess situation but are faster than novices at choosing a course of action.

Like anyone, experts store memories of different situations they've experienced but expertise lies in the development of a complex memory model that assists in assessing the situation at hand and contains knowledge of how rules should be applied (Lipshitz *et al.*, 2001). Decision making for experts who have developed these complex memory structures is less cognitively demanding; it is analogous to a learner driver who has to think consciously to decide when to cross a junction, change gear, brake, indicate, and so on whereas an experienced 'expert' driver finds these decisions so undemanding, they can hold a conversation and light a cigarette at the same time.

Of course, this type of multiple decision making relies on expertise and wide ranging experience. Less experienced managers of critical incidents who do not have a complex memory structure containing a broad repertoire of typical situations from which to generate first and best options are more likely to be prone to error. Expertise is clearly crucially

important when managing critical incidents.² Whitfield and Alison (2005) highlighted the many cognitive and interpersonal skills required by police critical incident managers, from making good decisions that will drive an inquiry forward, through managing the team and all the roles and responsibilities of each member, to managing relationships with communities and the media. Crego and Alison (2004) note the concerns of the Policing Skills and Standards Organization regarding the lack of such expertise. It is predicted that by 2010, 80 % of officers running major enquiries will have fewer than five years' experience.

However, experts can also be prone to error if the heuristics they rely on are too rigid and inflexible – in other words, if their situation assessment is wrong. To use a more concrete example, it is possible that the tragic events at Hillsborough in which 97 Liverpool football fans were crushed to death in an overcrowded pen, could have been compounded by RPDM. If officers were primed from past experience to recognize the unfolding but still uncertain situation such as a pitch invasion, they would have assessed the situation as a public order problem; in this case, the first option generated would have been to contain and control the crowd, which was the reported experience of many of those trapped in the pens. Recognized as situation A (public order problem), action X (contain the crowd) followed. Once the situation became clearer and it was recognized as a situation in which people were being injured, it was recognized as a situation of public safety rather than public order and police took action accordingly. Fortunately, the civil law recognizes that acting in an emergency, or crisis, is different. The standard of care expected is not so high, but that depends on the now past situation being acknowledged as having been an emergency. After the event, with the alleged benefits of hindsight, it may be difficult to understand how the police officers could have thought it was a public order event. This difficulty in understanding neatly reveals another cognitive bias that is empirically well established in the decision-making literature – the hindsight bias. Once we know the outcome of an event, we are more likely to regard it as having been foreseeable (Fischhoff, 1975). When making a judgement, individuals tend not to discount information that could only have been known after the event; as a result, antecedent decisions are more likely to be judged as decision errors *because* the outcome is known.

Decision making in critical incidents is a complex, dynamic and uncertain world with high stakes; of course, this complexity makes it particularly difficult to research. By retaining the factors that exist in the real-world beyond the lab (time pressure, uncertainty, stress generated by the situation, etc.), NDM has offered insights into what actually happens, though the relative contribution of each of these factors also needs to be understood. A balance needs to be drawn between preserving the rich, real context in which decisions are made and a more detailed, specific analysis of the relevant contributory factors. In the classic example used by gestalt theorists, we may wish to take the watch apart to understand how it works but all we are left with is a heap of wheels and cogs; it is no longer a watch.

Pragmatism

Pragmatism (Fishman, 1999) is an eclectic approach that can draw on the strengths of different decision-making theories. In line with NDM, it follows the principle of looking to

² This may be expected to transfer to other domains. For example, faced with unexpected evidence under time pressure, an experienced advocate could draw on a repertoire of typical cases and immediately generate a best option.

experts in a particular context as a starting point for research. Pragmatism is unlike TDM in this respect. TDM research begins with a theory from which testable hypotheses are derived whereas pragmatism views theory building as a secondary activity. Note, it does not eschew theory or model building completely. Its focus is on applied goals rather than theory for its own sake.

Having identified key issues of concern to practitioners in the field, and conducted qualitative research to identify themes relevant to the decision-making process, more traditional empirical studies can then be undertaken. In this respect, pragmatism retains the strengths of TDM and can isolate and control specific variables that may have an influence on decision making in the field. In a final process of triangulation, research results can then be fed back into the specific context from which the research question originated. This approach offers the opportunity for new learning to take place, for research to have a genuine applied value and retain scientific rigour.

The fact that pragmatic research remains rooted in the domain from which it emerged may be at the cost of generalizability. However, this is not regarded as a weakness. Decision making in critical incidents is a complex process in a dynamic environment and decision consequences can be profound. It is doubtful that a theory broad enough to transfer across domains would be rich or detailed enough to capture the specific and often unique nature of this domain. Lord and Hall (2005) argue that expert knowledge is organized around general principles (rather than surface features), it is domain specific and should not be expected to transfer at all to domains that do not build on the same principles. One size does not fit all and to attempt to impose more general theories on critical incidents may be to miss crucial and specific detail in the process (akin to building a theory of how *all* watches work). It is equally important not to narrow the focus to the other extreme. Crego and Alison (2004) warn against the danger of an atomistic approach whereby the complexity and interplay between different influences in a critical incident are lost (akin to focusing only on the heap of separate wheels and cogs).

In this respect, the eclectic nature of pragmatism can bring the flexibility required to accommodate the specialist nature of critical incident management without missing the sometimes unique detail, whilst preserving the 'big picture' that is meaningful to professionals. Critical incidents are powerful enough events that they inform changes in future policy making and can radically reshape political agendas (e.g. Macpherson, 1999; Bichard, 2004) so rigorous research in this area is crucial. Pragmatism's ability to embrace and incorporate more traditional experimental research work might also help to avoid the criticism sometimes levelled at NDM; namely, that NDM can be perceived as a poor relation if it does not embrace rigorous empirical work.

PRAGMATISM AND 10 000 VOLTS (10 kV)

A representative method of investigation within the pragmatic paradigm is Crego's (2002) ten thousand volts (10 kV) method (so called because the impact of a critical incident on professionals and the community has been likened to experiencing a 10 000 volt shock). The 10 kV is an electronic focus group originally modelled on learning systems within the aviation industry. Such systems provide a means of nonattributable reporting (to ensure full, frank and blameless reporting of incidents that endangered public safety). They also accumulate into a rich database of expert knowledge that can be accessed for research

purposes or future training needs; in short, such resources provide a practical means to ensure that the clichéd phrase 'Lessons will be learned' actually becomes a reality rather than just a media briefing soundbite.

Used as a debriefing method for senior police officers and other professionals, 10 kV elicits and stores the decision-making experiences from a wide range of enquiries, including critical incidents. In NDM terms, 10 kV begins by capturing 'what *was* done' and overcomes the prescriptive 'what *ought* to be done' emphasis of TDM. Participants record information on a laptop and their anonymized comments are shared with the whole group. Thus, a group discussion can be captured without the usual disadvantages of a public arena (where 'loud voices' dominate discussion and time limitations mean many views go unheard). The simultaneous recording in 10 kV sessions facilitates rapid and fully inclusive sharing of information whilst preserving the dynamic interactions of group discussion.

Example 10 kV Session

Crego and Alison (2004) conducted a 10 kV session involving 28 critical incident managers who had worked on cases ranging from sieges to child murders and the group generated over 250 statements in the first 20 minutes. Officers in the session indicated that one of the most significant issues for them in managing a critical incident was the way the police service would be judged by the wider community, victims and the media. It was also perceived as one of the most difficult aspects of an inquiry to manage. For example, delegates in that session commented on the difficulty of 'satisfying the insatiable appetite of the media and preventing their intrusion when they hold little to no respect for their voluntary regulatory authorities and attendant rules' (Crego and Alison, 2004, p. 11)

The issues that have an impact on decision making have been identified by the professionals themselves; not only does this have ecological validity, it also ensures that professionals regard any ensuing research as meaningful and credible. Having identified the research agenda, models and even theories may be developed thereafter but pragmatism's principle of utility means that a specific end goal always takes precedence over theory building.

FACTORS THAT INFLUENCE AN INDIVIDUAL'S DECISION MAKING

There has been much research into factors that affect decision making, both from TDM and NDM. Earlier research focused on cognitive processes of the individual decision maker and the heuristics and biases that influence the individual are well established empirically. Given that TDM regarded human decision makers as rational beings, emotion was not considered to affect decision making (these are also known as consequentialist theories). The impact of emotion on decision making emerged more from NDM research, along with other factors that influence decision making such as stress or time pressure.

In the real life situation of a critical incident, these factors will interact with each other. Time pressure often means that information is incomplete, ambiguous and uncertain; these factors can create or add to stress. In turn, stress may mean that information is ignored and thus incomplete. Given limited space, this section only looks very briefly at these factors

and how they can influence decision making or cause decision errors, but we shall return to the idea of emotion in the later section on decision avoidance.

Emotion

TDM models that focused exclusively on rationality neglected to consider the influence of emotions on decision making. Mellers, Schwartz and Cook (1998) offer a substantial review of the factors that can affect decision making and the interested reader may find it a useful source for further study. Some of the major findings are outlined below. Isen (1993, cited in Mellers, Schwartz and Cook, 1998) established that positive emotions increase creative problem solving. Estrada, Isen and Young (1994) found that decision makers with positive emotions integrate information more efficiently than controls. On the other hand, Luce, Bettman and Payne (1997) found that negative emotions can lead to faster use of available information and this can lead to increased choice accuracy in some situations. However, increased accuracy only obtains in situations where the task is easy. If the task is hard, accuracy is decreased when the decision maker experiences negative emotions. Critical incident situations might be intuitively characterized as situations where the task is hard, but, again, experts' perceptions of the task need to be considered so experience may moderate the effect here.

The emotions of anticipated regret or blame can also influence decision choices in different situations (See Bar-Hillel and Neter, 1996, on gambles; Simonson, 1992, on consumer products; Richard, van der Pligt and de Vries, 1996, on sexual practices). These emotions will be discussed further in the section on decision avoidance.

The empirical findings above are largely from lab studies and may or may not transfer to the domain of critical incidents though they give a general flavour of how emotion can affect decision making. In addition, the focus is on individual processes and although Josephs *et al.* (1992) found that feedback on decision outcomes can influence anticipated regret, the research does not fully address the influence of broader contexts.

Stress

Stress is a ubiquitous everyday term and we must make some distinctions; a feature of the situation that may cause stress is termed the stressor and stress is the individual's response. It may be physiological (pounding heart, sweating, etc.) or psychological (fear, worry, etc.) or behavioural (e.g. reactive depression, post-traumatic stress disorder). Individual differences exist so an objective measure of stress is unhelpful. Some may enjoy the thrill of a roller coaster whilst others will find it stressful and frightening, so definitions must incorporate the individual's cognitive interpretation. Jones and Bright (2001) define stress in terms of the relationship between an individual and the environment, which is perceived by the individual as taxing or exceeding their resources to cope and endangering their well-being.

Stress can impair cognitive processing. Research on attention shows a decline in cognitive performance even when simple dual tasks are undertaken (Pashler and Johnston, 1998) and critical incident management entails many simultaneous complex tasks. Stress can further narrow attention (Kahneman, 1973). More generally, Fiedler (1988, cited in Mellers,

Schwartz and Cook, 1998) found that any negative affect narrows attention and can result in a failure to search for new alternatives. So it may be ironic that decision makers, anxious about doing a good job in order to avoid legal liability, perform less well.

Thus, decision makers under stress can miss or ignore vital information, a condition termed premature closure. A practical remedy might therefore be to make decision makers aware of and guard against such possibilities by being vigilant. Janis, (1982, cited in Keinan and Friedland, 1987) argued that vigilance could contribute to rational decision making by ensuring a complete and thorough search for information so that all options are generated. However, in stressful situations, this can be replaced by hypervigilance where the individual may consider information in a very disorganized fashion and make poor decisions.³

Of course, these findings are interpreted within the paradigm of TDM whereby a failure to conduct a complete information search is regarded as a negative decision strategy and it is not necessarily relevant in naturalistic situations. Klein's (1993) RPDM model showed how experts use a very different decision strategy that does not rely on generation of all options and explains how stressful situations are managed. However, this focus is very much on the here and now of the decision situation.

In the broader context, role conflict and role ambiguity have been identified as significant causes of work-related stress; role conflict refers to the difficulty in managing competing demands from different individuals or groups, and role ambiguity is caused by uncertainty about what the job entails. Work overload is also a significant factor (Wastell and Newman, 1993). We can see the relevance of all these factors to decision making in critical incidents. Many groups demand different information (e.g. the media, senior managers, other agencies). A less experienced professional may well be uncertain about what the job entails and, moreover, has less recourse to RPDM decision strategies that rely on expertise and may buffer against the effects of stress.

Certainly, stress is an important area of research for professionals involved in critical incidents as repeated exposure to stressful situations puts these workers at greater risk of suffering stress responses or even being diagnosed with post-traumatic stress disorder (Everly and Mitchell, 1992, cited in Kowalski, 1995). Doran and Alison (2005) argue that more emphasis is needed on the relationship between stressors and stress moderators. Patterson (2003) found that seeking social support helped to buffer the effects of work-related stress amongst police officers but worryingly, Pogrebin and Poole (1991) found that US police officers regard the discussion of emotions and stress responses as taboo because they may be viewed as a personal inadequacy; it is therefore rarely discussed. This is an example of how organizational norms can affect individuals (of which more later) and in a reciprocal fashion, how negative emotions that are unresolved may affect future decision making.

Ambiguity

Information and knowledge in critical incidents is typically uncertain. It may be incomplete, or overwhelming in quantity and subsequent impact. Both can lead to ambiguity. Frisch and

³ A practical application of decision-making research can be made here. An individual is innocently caught up in an affray; a third party offers assistance; stress narrows attention so not all available information is processed; the individual perceives the helper as a threat and *decides* to defend themselves. Thus, the erstwhile 'victim' ends up facing an assault charge.

Baron define ambiguity as 'The subjective experience of missing information relevant to a prediction' (1988, p. 152), so missing pieces of vital information can make the situation ambiguous. Information overload means reliance on selection and interpretation, which can also constitute ambiguity if the relevant information is not selected or not understood. TDM studies have shown that decision makers are averse to ambiguity and prefer unambiguous options (Ellsberg, 1961, cited in van Dijk and Zeelenberg, 2003). However, in naturalistic environments, an unambiguous option is often not available and decision makers must contend with uncertainty. Van Dijk and Zeelenberg (2003) found that decision makers simply discount ambiguous information. In the context of a real time and still unfolding situation, the discounted information may transpire to be crucial.

Aversion to ambiguity may drive a decision maker in the opposite direction; rather than discount ambiguous information, they may embellish it. Decision makers may generate a narrative or story to explain causes and then base subsequent decisions on their explanations (Pennington and Hastie, 1992). Further, decision makers are likely to infer meaning in the absence of solid information (Innes, 2002). For example, the absence of information or an explanation can make evidence, in court, difficult to believe. Evidence in court can also be difficult for lay jurors to understand and, hence, a story 'template' assists them in categorizing, organizing and understanding complex evidence. Such templates also have disadvantages. Mock jurors were found to infer details that had not actually been contained in direct testimony; they also systematically deleted evidence that was unrelated to their preconstructed narrative of events. The decision-making concern here is that jurors are thought to use the stories they have constructed to decide on verdicts (Pennington and Hastie, 1986).

Decision makers, then, may extrapolate from what is certain (known as assumption based reasoning) to reduce uncertainty (Lipshitz and Strauss, 1997). They may also forestall, which is a form of decision avoidance. Whilst such strategies for dealing with uncertainty may be helpful, without ongoing review, the strategies may result in decision errors.

In naturalistic situations, this potential for decision errors may be compounded by other factors. Morrison *et al.* (n.d) studied tactical decision making in the navy, where participants had to track visual computer generated displays (known as geo-plots) and identify weapon threats. They found increased hypervigilance (and the attendant attentional narrowing) and increased confirmation bias when information was ambiguous.

Time Pressure

Ariely and Zakay (2001) argue that decision makers can and do adapt to time pressure. They may respond by being more selective about which information to consider when weighing options but under intense time pressure, they are likely to switch decision strategies. Klein (1993) found that RPD decision strategies can be used to good effect under time pressure, but Kahneman and Tversky (1973) found that a greater reliance on heuristics may lead to greater confirmation biases. Decision making may even become dysfunctional. Dörner (1987, cited in Wastell and Newman, 1993) proposed the term 'Intellectual Emergency Reaction' whereby individuals in critical incidents became more and more rigid in their thinking and decision making, focusing on narrower issues and selecting only options that confirm their rigid heuristics. The dangers of simplistic pattern matching coupled

with a lack of evaluation and review of actions and the contribution to decision errors are clear.

Although considered separately here, factors that influence decision making may be co-present and, moreover, interact in a critical incident situation to influence decision making. Whilst experience can help decision makers to deal with situations, expertise is no simple panacea that will ensure good decision making. Indeed, greater experience means greater exposure to the broad context; that is, the organization, which may be a significant factor in decision avoidance. We shall discuss both in the next section.

DECISION AVOIDANCE

Everyone can readily call to mind examples where failure to make a decision and take subsequent action has resulted in tragedy. Anyone who has read a newspaper or watched television knows that the one of the first questions asked by news journalists is: 'Why did the relevant authorities not act sooner?' when an abused child has not been taken into care or a murderer turns out to have had a lengthy history of preconvictions for violence. The failure to make decisions has also been a cause for criticism in public inquiries into critical incident management (e.g. Laming, 2003). Yet despite the ubiquity of such tragic stories and the prominence of awareness amongst the general public about the failure to make decisions and take subsequent action, there has been surprisingly little psychological research into why people fail to make decisions. The focus is overwhelmingly on how and why decisions *are* made.

A similar position may be noted with regard to the law. People can be prosecuted or sued for harm caused by their omission to act, not just for acts of commission. There has to be a special duty of care (e.g. a very close family relationship or the individual was responsible for the danger which they failed to avert) for criminal liability. However, the focus, both in civil and criminal law and practice, is overwhelmingly on positive acts. A partial explanation is that we do not as readily 'see' the consequences of an omission to act as we do a commission; that is, they are not as salient. We know the consequences if a doctor gives us the wrong medication (well sometimes), but we are unlikely, without special skills and knowledge, to know the consequences of the doctor's failure to provide us with a different treatment.

Anderson (2003) attributes some of this failure to focus on decision avoidance to the many subdisciplines within psychology and the constraints of particular paradigms used by researchers. If Anderson's claim is true that 'decision avoidance is the dominant attitude of human decision makers within certain boundary conditions' (2003, p. 140), this is an extremely important omission in the literature; pragmatism's eclecticism may provide some opportunity to redress this narrow focus and investigate the phenomenon of decision avoidance.

Anderson defines decision avoidance thus: 'A tendency to avoid making a choice by postponing it or by seeking an easy way out that involves no action or no change' (2003, p. 139). It is distinguished from procrastination whereby an individual has a specific intention. We are all familiar with our decisions to write the essay, pay the bills, visit an ageing relative; we have, in effect, *decided* on a particular action but don't carry it through usually because we'd rather watch television or go to the pub.

Anderson's (2003) Model of Decision Avoidance

Anderson proposes an integrated model of decision avoidance that incorporates both rational and emotional influences on decision avoidance and he cites selection difficulty as the major contributor to decision avoidance. There are many factors under the umbrella of selection difficulty in Anderson's model; we shall consider two here – blame and regret as contributors to decision avoidance. Anticipated blame or regret are construed as mental simulations of future outcomes. Anticipatory blame or regret are felt during the decision-making process and individuals may seek, in various ways, to reduce these uncomfortable feelings. Indeed, for Anderson, managing emotions underpins decision avoidance.

Laboratory studies show support for the idea that anticipated regret leads to decision avoidance in the form of omission bias (discussed below). Ritov and Baron (1990) argued that anticipated regret explained the result that participants refused a (hypothetical) vaccination despite the fact that the fatal side effects of the vaccine were a fraction of the death rate posed by the disease. This is clearly not rational in the strict terms of TDM theory. One *ought* to consider all the information and choose the option that offers a substantially lower chance of dying. However, the regret anticipated by positively, proactively choosing the vaccine option was far greater than omitting to choose it and thereby avoiding responsibility for the decision. (See also Baron, 1991; Baron and Ritov, 1994; Spranca, Minsk and Baron, 1991.)

Anderson identifies four effects as follows: (1) status quo bias, (2) omission bias, (3) choice deferral and (4) inaction inertia (although inaction inertia is seen to overlap with the other three factors).

Status quo bias – whereby an individual prefers the status quo and will tend to avoid decisions that may change it: The status quo is the normal state of affairs and deviation from the status quo is, therefore, seen as abnormal. Norms are highly salient and considering deviation from them will give rise to uncomfortable emotions that must be managed. For example, if a social worker considers the decision to take a child at risk into care, they might anticipate the blame and regret that that will ensue if they make the wrong decision (by imagining or mentally simulating the outcome). These uncomfortable feelings of anticipatory regret or blame may increase their propensity to retain the status quo; that is, leave the child at home where they are at risk of significant harm. Organizational context can also have an influence (this will be discussed in more detail in a later section). If the social worker works in an organization that has a policy of keeping children with their families wherever possible, these organizational norms will also be highly salient and a deviation from the organizational status quo will also be perceived as abnormal and may contribute to decision avoidance. On this view, retaining the status quo can reduce an uncomfortable regret for the decision maker. (Of course, this obtains *during* the decision-making process. Regret would be severe in the event of a tragic outcome.)

Omission bias – whereby individuals prefer options that entail no action: There is empirical evidence from decision-making literature that acts of commission are perceived as more culpable than acts of omission (Spranca, Minsk and Baron, 1991) and Kahneman (1995, cited in Mellers, Schwartz and Cook, 1998) showed that action is perceived as leading to more regret than inaction. This too may contribute to decision avoidance; less blame or regret will accrue from doing nothing than doing something, so if outcomes are uncertain, it is better to err on the side of caution in a bid to avoid a decision error.

Choice deferral—here individuals choose to delay making a decision: From a TDM perspective, this may be construed as a positive decision strategy to wait for more information and conduct a thorough search but if time pressure of the situational context dictates that a decision must be made quickly, it is construed here as decision avoidance.

Incidentally, this highlights the importance of considering context in decision making and some of the difficulty in interpreting decision strategies. What is positive in a context of unlimited time (waiting for more information to clarify the options) is negative in a different context (harm may have resulted by deferral).

Dhar (1997) found that choice deferral decreased under time pressure. A possible explanation for this effect lies in the decision-making strategy adopted. Recall that Klein (1993) found support for his RPDM model amongst fire commanders under time pressure. If only one option is generated, selection difficulty is absent; if a decision maker does not weigh the values of different options, anticipatory regret or blame will not be elicited and a decision can readily be made rather than avoided.

However, this ignores the impact of organizational context and culture; it is also possible that the mental models employed during RPDM contain prior experience of blame or regret that can affect the decision-making process in either direction—to decide and act or to avoid. For example, recent experience of a public enquiry that named and shamed individuals within one's own organization or profession might well elicit mental simulations of anticipated blame or regret and evoke anticipatory blame or regret that might lead to decision avoidance; in the pattern matching terms of the RPDM model—if situation A, then action 0 (nothing at all if you want to avoid blame). There may be long-term implications here for the decision-making abilities of people whose organizational context is one where a blame culture is chronic and/or endemic.

Essentially, there are two different but relevant trajectories here, the immediate (proximal) context and the long-term (distal) context. If time pressure is a feature in the proximal context of a critical incident, less decision avoidance should be predicted. If a name, shame and blame culture is a feature of the distal context, more decision avoidance may be predicted. Both proximal and distal contexts can influence decision making and the next section will focus on theories that might offer ways to determine the relative impact of each.

ORGANIZATIONAL CONTEXT AND CULTURE: IMPACT ON DECISION MAKING AND DECISION AVOIDANCE

Social Identity Theory and Self-Categorization Theory

Social identity theory (SIT) (Tajfel and Turner, 1986) and its subsequent development, self-categorization theory (SCT) (Turner *et al.*, 1987) may offer valuable insights into the impact of organizations on the behaviour of individuals, including decision-making behaviour. Social identity is defined as 'An individual's knowledge that he or she belongs to certain social groups together with some emotional and value significance to him or her of this group membership' (Tajfel, 1972, p. 31, cited in Haslam, Postmes and Ellemers, 2003), for example one may belong to the social groups, student, Manchester United fan, and so on. Self-categorization theory extended the idea of social identity and is concerned with an individual's subjective sense of self; SCT offers a continuum between individual

and collective identities that become increasingly inclusive. At one end of the continuum, any one person has a sense of themselves as an individual with their own unique identity; at the other extreme, we all share a collective identity as a human being.

For our purposes here (in the middle of the continuum), individuals will self-categorize and may have a social identity relating to their work organization. An individual knows they belong to a particular professional group and it is meaningful to the individual to define themselves as a police officer, a lawyer or a social worker. Importantly, social identity has an impact on behaviour. It would be unremarkable to find that a police officer is less likely than a career criminal to rob banks regularly. Moreover, the theory predicts that people with the same social identities will perceive their views as interchangeable. A Liverpool fan would expect another Liverpool fan to dislike Manchester United supporters because it is one of the defining features of the social identity 'Liverpool Fan' that they do not like (and, crucially, are not like) Manchester United fans. There are, of course, many individual fans that are not so extreme in their personal views and herein lies the influence of groups and social identity.

Social groups have norms and any individual must negotiate conflict between individual identity (e.g. uncomfortable prejudging all Manchester United fans) and a particular social identity (e.g. pressure to conform to group norm of rivalry with Manchester United fans). Strong identification with the group should predict behaviour in accordance with group norms.

Group norms may also underpin the idea of shared mental models (defined as individuals who possess a similar cognitive representation of a situation or phenomenon) (Langan-Fox *et al.*, 2001). A shared mental model is thought to enhance effectiveness of teams working towards a shared aim or goal. Discussion of teams is precluded here (see Flin, 1996, on teams in critical incident management), but an individual's social identity and self-categorization as part of a team will be mediated through identification with the larger group; that is, the organization, and we may look to those organizational norms to identify the derivation of mental models or a 'team mind'.

Organizational Identity

Social identity and self-categorization theories, then, offer an explanation for the influence of group norms on behavioural outcomes. This relates to the individual decision maker, thus: the stronger the identification with the organization, the more the decisions will be influenced by the prevailing organizational culture. Haslam, Postmes and Ellemers (2003) identify three factors that are likely to strengthen an individual's organizational identity as follows: (1) the length of time they have worked there; (2) whether the organization is in competition with another organization; and (3) if they feel proud of the organization.

We can consider our earlier hypothetical example, the social worker who does not take the child into care although they are at risk of significant harm. Anderson's (2003) model of decision avoidance is helpful in explaining these processes in the individual: the worker displays a status quo bias. The worker may experience anticipatory regret in the decision situation. In order to manage this uncomfortable anticipatory regret they feel, the worker avoids a decision and sticks with the status quo. The worker may also experience anticipated blame or regret whereby they construct mental simulations of future outcomes.

These contributory factors are features of the proximal context but, crucially, they do not emerge from nowhere. Such ideas and feelings must have an earlier source.

In SIT/SCT terms, the distal context influences the proximal context; put simply, the past will influence the present and the decision maker's mental simulations of future outcomes will be based on prior experience (a factor also identified by Klein and incorporated into his RPDM model). SIT/SCT can help to explain the mechanisms through which these earlier sources (the distal context) can evoke such uncomfortable emotions in the decision-making situation (the proximal context).

Consider the recent political culture that promotes accountability in public sector organizations such as the police service. This generates a focus on accountability in each force (often framed in terms of audited targets and performance indicators). This context will have a cumulative impact on the organization's identity. Accountability is high on the agenda, is increasingly salient and becomes one of the norms of the organization. This will have an impact on the organizational identity of its employees; to self-categorize and identify yourself as a police officer, you will be aware that you are a public servant and conform to the organizational norm of public accountability. If this accountability has manifested as an experience of public blaming and shaming, an individual decision maker will carry that experience forward; it is an integral part of their organizational identity and can influence their decision making. This is the route through which heuristics and biases may be formed. Decision makers will anticipate blame or regret for decisions made (proximal context) if that is the history of the organization with which they identify (distal context).

Recall that earlier we mentioned that delegates in Crego and Alison's (2004) 10 kV session identified such public accountability as one of the most significant issues in managing critical incidents (i.e. how they would be perceived by the community, victims and the media). They also found it one of the most difficult aspects to manage. It is easy to see the potential for long-term damage to decision-making skills in an organization that has experienced public blaming and shaming. In terms of Anderson's (2003) model, anticipated blame or regret is a major contributor to decision avoidance.

Recall too that an individual's organizational identity will be strengthened if they feel proud of the organization and/or perceive it to be in competition with others (Haslam, Postmes and Ellemers, 2003). We can see how a defensive culture emerges when an organizational identity contains (very salient) notions of anticipated blame for decisions made. Decision makers are likely to be more concerned with sticking together (if they are proud of and identify with their organization) and watching their backs (if outsiders are regarded as the competition) than they are about making decisions.

Again, illustrated with the earlier example from Crego and Alison's (2004) study where critical incident managers found the media difficult to manage, particularly given that the intrusion was perceived as contravening regulatory codes of practice. The media represent outside organizations (with different identities from the police). Moreover, they may be perceived as competing (for access to and control of information). This will strengthen the organizational identity of individuals in both the police and the media and a concomitant effect in the long term may be to increase the difference in identities between the two organizations. For the individual, it is extremely difficult to manage someone who is adhering to organizational norms that are very different from your own. They are unfamiliar or even unknown (the police's only reference point may be regulatory codes of practice for the media; they may not be organizational norms in reality). The ensuing uncertainty may cause anxiety and affect decision making or tend towards decision avoidance if blame or regret

is anticipated. People are entitled to complain; indeed complaints can be a valuable source of information, but many complaints procedures then provide for the person complained against to be suspended from work. However technically proper such procedures may be, and however often it may be stressed that the complaint is not being prejudged, they rarely convince, especially when the person criticized finds it more difficult to mount a defence to the accusation.

Abstract Communication Systems Theory

There are some parallels between SIT/SCT and abstract communication systems theory (Luhmann 1996, cited in Andras and Charlton, 2005). Communication systems theory may offer insight into the processes of how organizational identity becomes established. A detailed exploration is precluded here but, essentially, the management of an organization has three main functions, all of which relate to organizational identity. They are: (1) to define the organizational identity, (2) to check the organizational identity and (3) to enforce the organizational identity. The communications or rules of the organization serve as the conduit and act as a reference point against which identity can be checked and confirmed or disconfirmed. Thus, a department that does not comply with the rules as communicated by the management would be closed down (and individuals would be absolutely aware that they are not conforming to the norms of the organizational identity).

In an applied sense, one can see the propensity for decision makers to stick closely to policies and procedures because they will act as a reference point that a decision maker is adhering to organizational norms. Existing policies and procedures are powerful communicators of the organizational identity; as such, they exert influence on decision makers not to stray from them. To do so would be to stray from the individual's own organizational identity and potentially evoke uncomfortable and dissonant feelings of, say, anticipatory or anticipated blame or regret.

MULTI-AGENCY PARTNERSHIPS AND INACTION INERTIA

Different (or worse still, competing) organizational identities may also militate against successful multi-agency partnerships. Smith and Dowell (2000) highlight the difficulties engendered by a lack of shared mental models across agencies. Hallett (1995, cited in McClean and Alison, 2005) commented that one of the most persistent findings of child abuse inquiries is the recurrent failures and weaknesses in interagency co-ordination. Sinclair and Bullock (2002, cited in McClean and Alison, 2005) analysed 40 UK child abuse case reviews and found inadequate sharing of information as one of the most common shortcomings.

In terms of communication systems theory, each organization will have different systems of communication or rules and the language of such communications is unlikely to translate readily across agencies. For example, a request to a police officer to 'Check the history of the case' is likely to result in a search of the Police National Computer (PNC) database for previous convictions; the same request verbatim to a social worker is likely to elicit sociological details of family background, living arrangements, and so on; a health worker may report past medical conditions. Murphy (2004, cited in McClean and Alison, 2005)

found evidence to support this idea of different languages and systems between agencies as a contributing factor to problems in multi-agency co-operation.

From a SIT/SCT perspective, individuals will be adhering to very different organizational norms. Reder and Duncan (2003, cited in McClean and Alison, 2005) found that a lack of trust between agencies posed a major difficulty. McClean and Alison (2005) analysed a 10 kV debrief following a multi-agency critical incident exercise (MACIE), involving professionals from health, education, social services and the police service who had collaborated on four simulated child abuse inquiries. Again, accountability emerged as a major concern for these public sector professionals. Decision avoidance was prominent in the form of omission bias. Having identified child protection concerns, little action ensued (in Anderson's terminology – inaction inertia). Moreover, this was often attributed to a failure of other agencies. Sample comments were:

Police and Social Services did not carry through the decision to do a home visit
Why did health not follow through decisions of strategy meeting? (2005, p. 12).

SIT/SCT and communication systems theory (through aspects of competing organizational identities and different communication systems manifested as different procedural priorities) can contribute to our understanding of how blame came to be a common feature in this study and help to explain the defensive stance assumed by several delegates.

In the private sector, competition is expected and regarded as healthy. One burger company does not expect its employees to collaborate with those of a different fast food organization so strengthening organizational identity via competition poses few practical problems. However, in the public sector, 'joined up thinking' is a stated aim of the government and multi-agency partnerships are an increasing reality in the current political climate. The chair of the public inquiry into the death of Victoria Climbié, Lord Laming, stated, 'The future lies with those managers who can demonstrate the capacity to work effectively across organizational boundaries (because) such boundaries will always exist . . . those who persist in making decisions alone . . . must either change or be replaced' (2003, p. 9). This constitutes a significant, powerful context for (and subsequent influence on decision making in) public sector organizations in the long term. We can also see some of the factors that make it difficult to achieve the aim of working together; that is, how different organizational identities between agencies that are expected to collaborate may affect decision making and increase decision avoidance.

Even for a single agency, policies and procedures may be unlikely to encompass enough information to assist decision makers in critical incidents because they are, by definition, atypical or even unique events. In the uncertain and complex situation of a critical incident, a decision maker may be 'creative' enough to generate a solution to a problem but we can also see some of the very powerful organizational influences that militate against an individual thinking and making decisions creatively and more importantly, taking action based on the decision. It would require a singularly robust individual to do so. In 2002, it was acknowledged that, 'there are very few experienced senior investigators within the police service who could easily manage challenges at this level' (Metropolitan Police Service, 2002, p. 14).

There is, then, pressure from organizational norms mediated through the individual's own social/organizational identity to stick to the rules, policies, procedures and keep to the

status quo. It is unsurprising, in typically stressful, uncertain situations that strong emotions are elicited and decision avoidance occurs.

Where decisions are made, decision errors may occur because options generated may not fit the unique features of a critical incident. Pressure to adhere to policies and procedures leaves little scope for an individual decision maker to reject their organizational identity and act as an individual who is (1) creative enough to generate a bespoke solution and (2) brave enough to act on that decision. It may be no coincidence that it is the rare or even unique incidents that ultimately result in public inquiries.

CONCLUSION

We have reviewed different theories of decision making and considered their strengths and weaknesses. We have given a necessarily brief outline of some of the factors that can influence decision making in different situations and begun to explore some of the broader theories that may help us to understand the context behind the complex nature of decision making in critical incidents.

In this chapter, the emphasis has been on problems that can occur (decision errors and decision avoidance) to the detriment of exploring solutions. As mentioned earlier, description is a useful start point but not the end point for research. In the absence of complete information in this chapter, the reader should not infer that psychologists have limited their research to identifying and describing the difficulties in decision making. Vigorous research is ongoing; solutions and models, theories and practical tools have been designed to support decision making. For example, Crego (2002) designed Hydra, a sophisticated high fidelity immersive system that simulates critical incidents in real time over hours and days. Devised for and used by the police service in the UK, Hydra is an unparalleled resource and is leading the world in training and preparing practitioners for response to crises. (See also Crego and Alison, 2004; Hutchins, 1996; Schraagen *et al.*, 2005; McCann *et al.*, 2000; Kontogiannis, 1996.)

It is also crucial to investigate success and not focus exclusively on failure in decision making. Such research has much to contribute in supporting critical incident managers who repeatedly deal with chaotic, complex and life-threatening situations. They continue to do so despite an ever intensifying blame culture and they are often overwhelmingly successful in their efforts. We should be wary of our own availability and hindsight biases; news stories of failure and tragedy are salient to us all.

A topical example is the London bombings in July 2005. Comparatively little attention has been paid to the professionals in the emergency services who managed and co-ordinated an enormously complex response to the tragedy as they rescued, treated and continue to treat scores of injured and distressed people. In the face of 'outrageous fortune', they decided to act 'against a sea of troubles' on behalf of us all.

POSTSCRIPT

The London Assembly report into the 2005 London bombings, prepared by Richard Barnes, has been published since this chapter was written. It contains a finding that may be pertinent to future decision avoidance.

At the end of the chapter, we mention that, in the midst of chaos and uncertainty in a critical incident, a very robust individual would be required to decide and act on a creative 'bespoke' solution if the decision deviates from policy and procedural norms. Just such a decision was made by James Hart, the Commissioner of the City of London Police.

The context was this: Under Access Overload Control arrangements (ACCOC), network providers can render the mobile phone system inaccessible to the vast majority of users, meaning that only selected individuals can use their mobile phones. Strategic 'gold' commanders in charge of the incident had decided not to activate ACCOC for fear of panicking the public and also because most Metropolitan police personnel did not possess ACCOC-enabled mobile phones.

Mid morning on July 7th, as mobile phone networks struggled under the volume of calls, James Hart was still unaware of the strategic decision *not* to activate ACCOC. He made a solo decision to activate ACCOC. In collaboration with a network provider, James Hart established a one kilometre 'exclusion zone' around Aldgate tube station. The Barnes report found that,

The City of London Police acted outside the [command and control] framework. This should not be allowed to happen again (2006, p. 48).

Thus, Hart may be seen to have been publicly censured for deviating from policy and procedure despite the important practical benefits of his decision; namely, he reported that, as communications deteriorated, his decision/action had effectively freed up the airwaves for emergency workers to communicate via mobiles between the scene at Aldgate and casualty reception at hospitals receiving the injured and, rather ironically, that a press liaison officer of the Metropolitan Police had borrowed ACCOC-enabled mobiles to communicate what was happening at the scene. Some of the difficulties inherent in multi-agency working are highlighted as well as the long-term implications. If Hart's colleagues were to find themselves in a similar situation in the future, the potential for decision avoidance is clear.

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